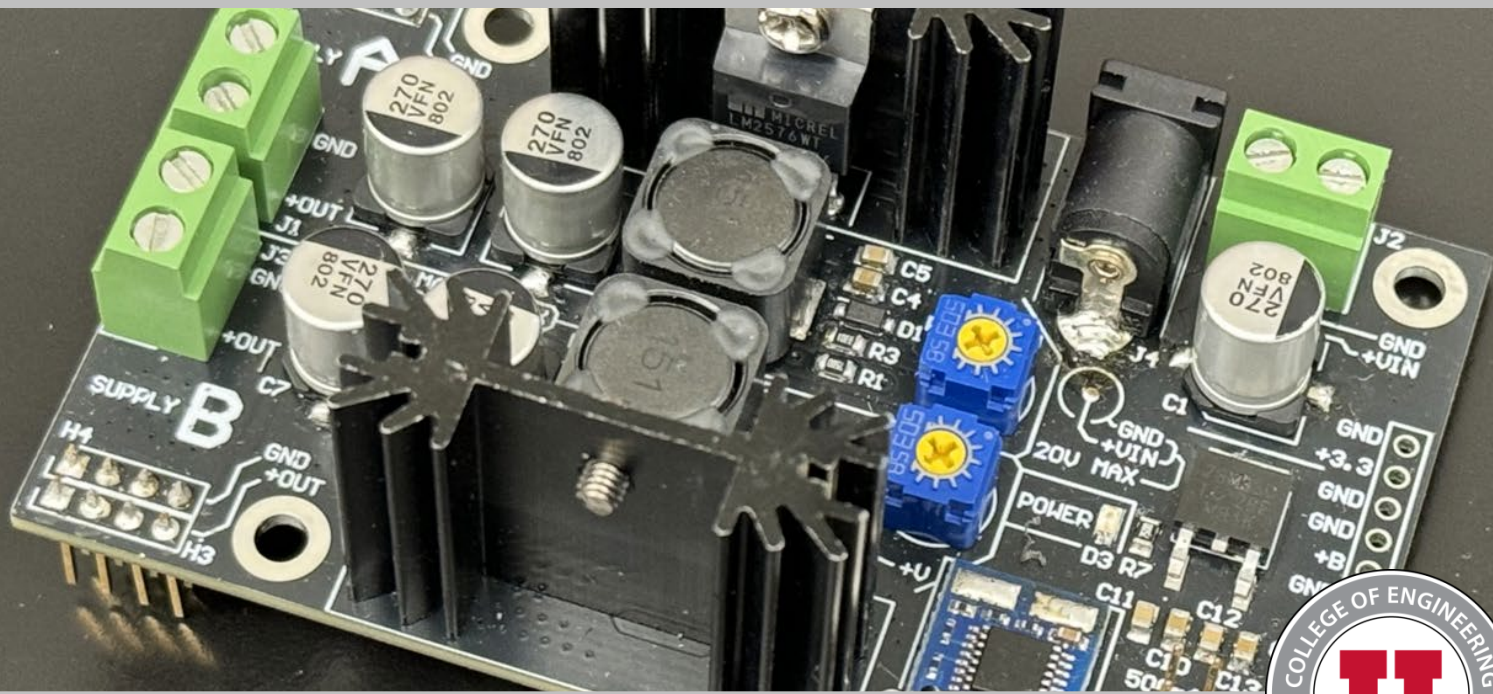


IEEE PCB Design Workshop: (Week 06) Introduction to Soldering



Hosted By: Adrian Sucahyo and IEEE at the University of Utah
Adapted From: IEEE x FSAE Workshop SP25 with Nick Howard
and Adrian Sucahyo

Workshop Outline

Tentative Schedule:

- Sept. 3 – Introduction to Schematics
- Sept. 10 – Schematics and Components
- Sept. 17 – Introduction to PCB Layout
- Sept. 24 – Footprints and PCB Layout
- Oct. 1 – Open Work Session
- ** FALL BREAK **
- Oct. 29 – Soldering Week 1
- Nov. 5 – Soldering Week 2
- Nov. 12 – Soldering Week 3

Announcements

- Scheduling
 - We will be having 2-3 more sessions to finish the soldering portion of the workshop.
 - If you cannot make a session but still want to receive a board, please let me know.

Want more experience?

- Consider joining the FSAE tractive team!
 - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
 - No experience required!



U of U FSAE Discord Link



Join the IEEE Discord

- If you haven't already, please join the IEEE Discord server for additional information and updates regarding this workshop



U of U IEEE Discord Link



Soldering Overview

- Soldering is simply attaching two metal surfaces with a filler metal (solder)
 - Provides both electrical and mechanical connection
- Typically, soldering is performed with a hot iron or hot air
 - Irons are typically used for through-hole components or large SMDs
 - Hot air is typically used for SMDs
- The choice to use hot air or an iron is highly situation-dependent
 - Iron applies local heat
 - Hot air heats a large area

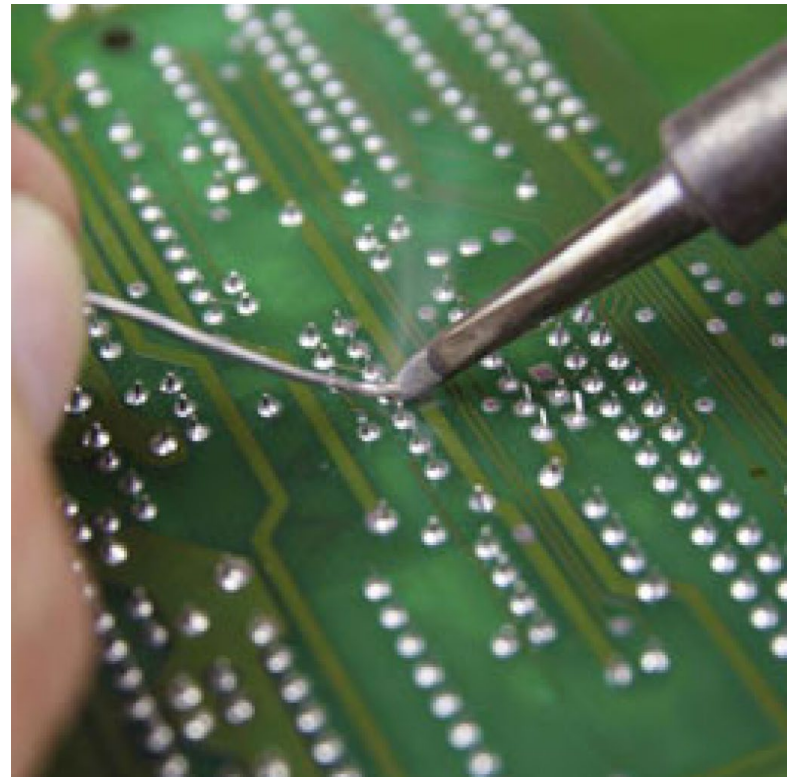


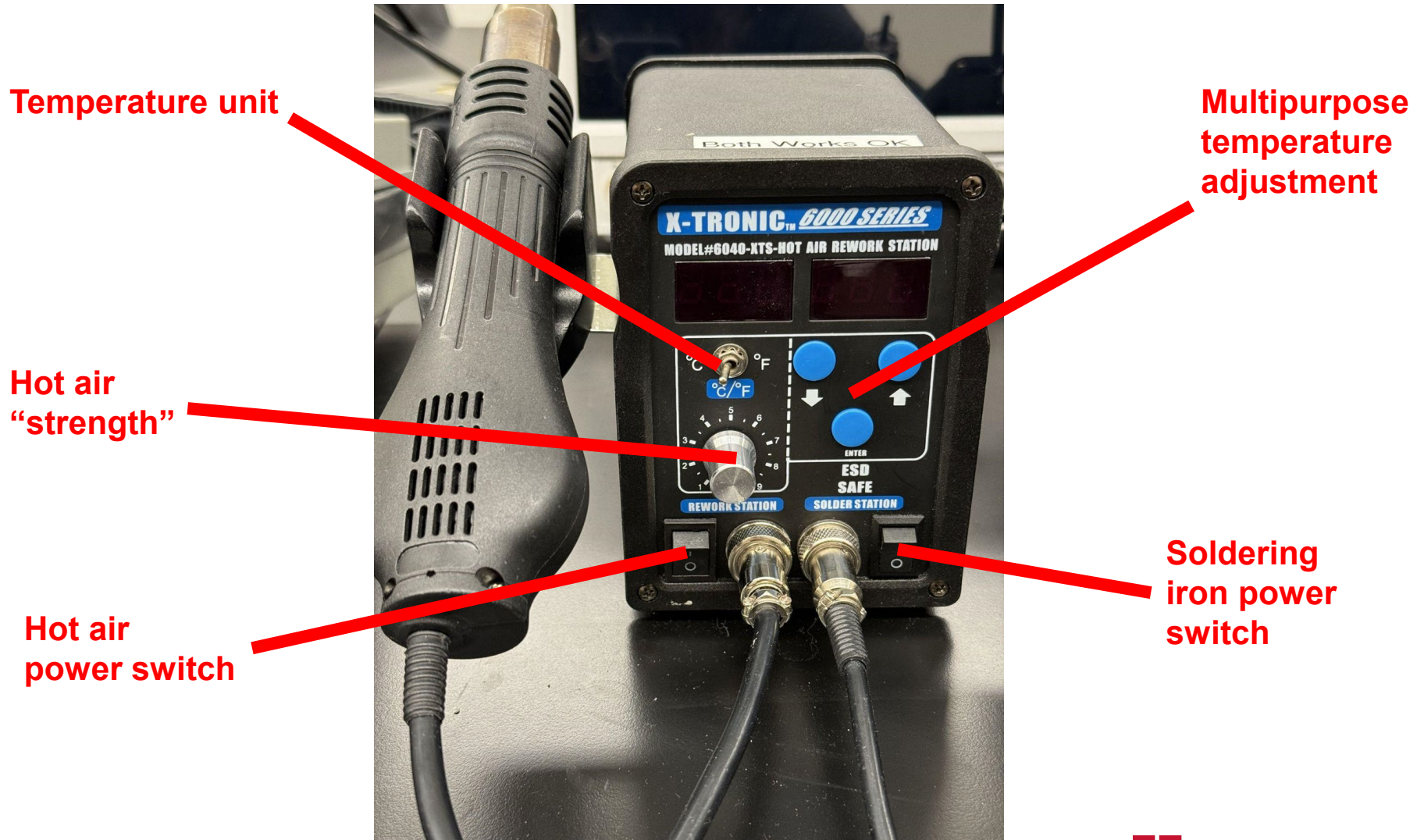
Image credit: RushPCB
<https://rushpcb.com/what-is-a-solder-joint/>

Solder Types

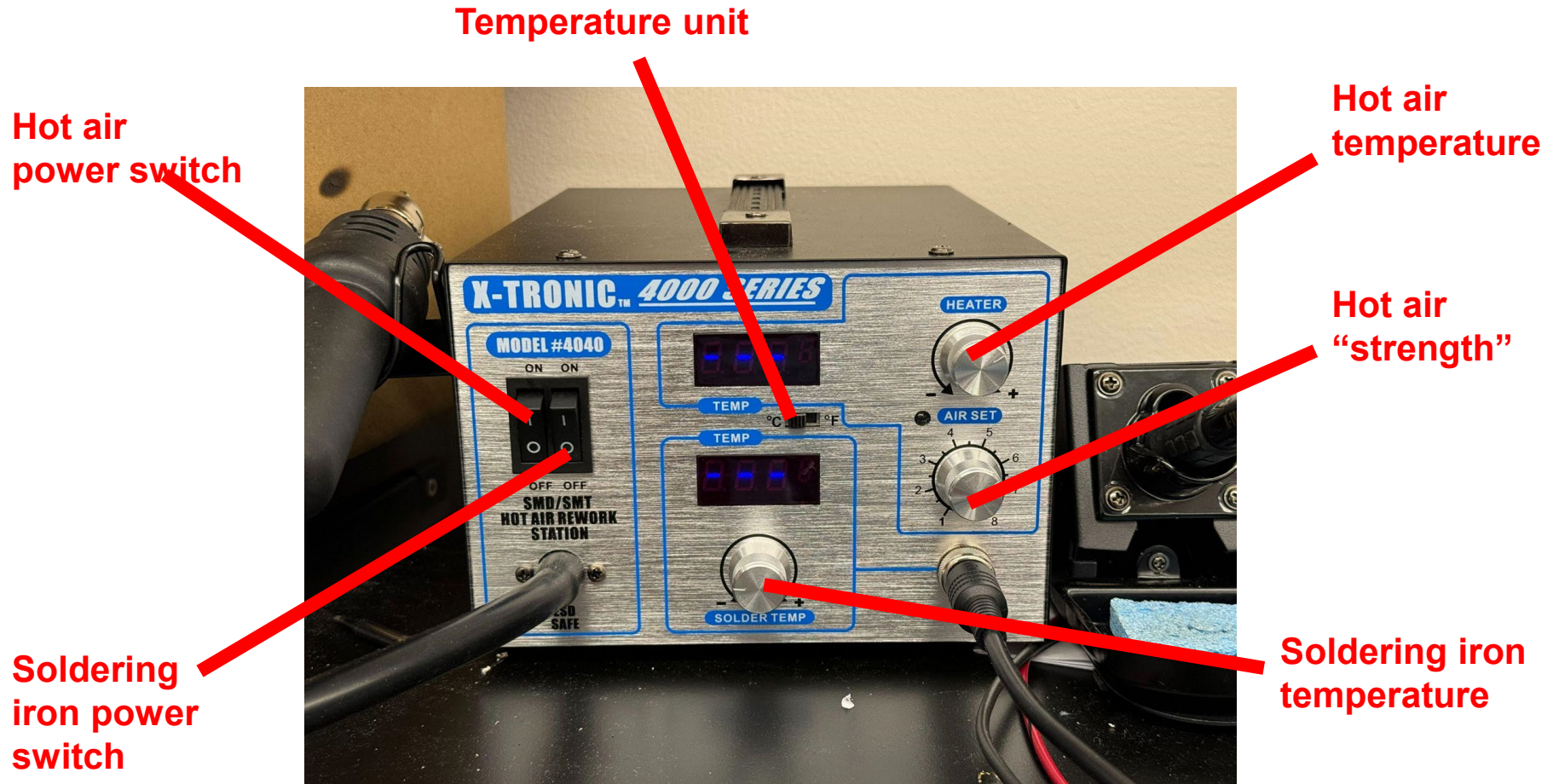
- Solder can have a variety of compositions
 - Can come as a spool (wire) or paste
- Pb (lead) solder is very easy to use
 - Relatively low melting point ($\sim 183^{\circ}\text{C}$)
 - Flows very easily
 - Typically, Sn 60% Pb 40%
- Pb-Free solder is a bit more difficult to work with
 - Typically used for RoHS compliance
 - Can be mostly tin (Sn 99% Cu 0.7% Ag 0.3%) or part Bi/In
 - Sn99 solder melts at $\sim 217^{\circ}\text{C}$
 - Bi/In eutectics can melt as low as $\sim 138/118^{\circ}$



Soldering / Rework Stations

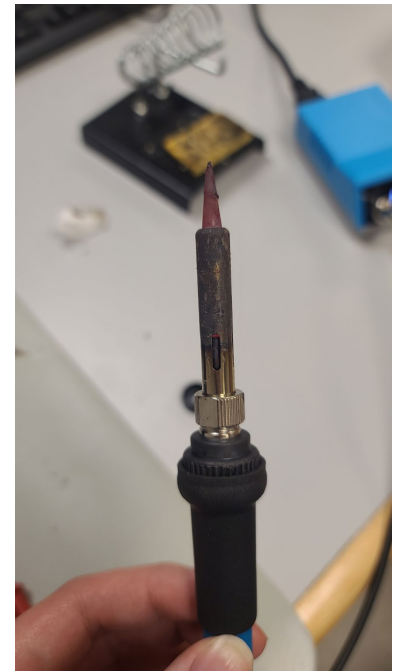


Soldering / Rework Stations



Important Notes

- NEVER touch tip of soldering iron or hot air gun while in use or shortly after use
- NEVER unplug hot air station while in use
 - Allow heating element to auto-cool when done
 - Turn off hot air with switch on front panel to auto-cool off
 - On 6000 series stations, gun will auto-cool off when placed back in holder
- Solder in a well-ventilated area
- Don't leave soldering iron or hot air gun on and unattended
 - Leaving soldering iron on for extended periods can damage tip
- Clean and tin soldering iron tip regularly
 - Tin iron before powering off
- Keep temperatures reasonable
 - Start low, increase as needed
 - Start Pb-free around 280-320 ° C



Questions?

Questions?

Workshop References

Navigate to the workshop GitHub and
download today's files

<https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Fall-2025>